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1- OVERVIEW

This procedure explains the functional checks from the RFI module for the 1.0T RF/PDU cabinet.

The RFI functional checks procedures described in this document should be performed when the RF/PDU Maximum Power RF Output setup and Calibration procedure has failed.

1-1 Function

The RFI module performs 2 major functions:

1. The RFI is the RF interface from the Transceiver Exciter Output to the RF power Modules, and finally to the RF Coils through the heliax cable connections.

   The RFI module splits the RF In signal from the exciter to provide RF Drive (Amp1 Drive and Amp2 Drive) to the 2 RF Power Modules in the RF/PDU cabinet. The RFI then combines the output of the 2 RF Power Modules in phase and provides the proper output power (via the gain adjustment pots) needed for Body or Head mode. The Head and Body Couplers, located in the RFI, provide the redundant pick-off signals to the Power Monitoring circuitry on the APM.

2. The RFI Module also provides communication translation between the RF Power Modules and the System Support Module (SSM) which in turn communicates with the host.

   The RFI functionality includes: Head/Body Switching, Test output, Serial Link decoding, error/status reporting, and unblank / RF Lock signal control.

Note

In the 1.0T RF/PDU cabinet the RF signal from the exciter is connected via a rear interface bracket to the RFI Module J14 BNC connector RF In (front). Envelope Feedback is not utilized. Therefore, there is no connection to J105 or J103 of the SSM.

2- TOOLS AND INSTRUMENTS REQUIRED

Refer to Table 2-1 for equipment required with the RF Power Measurement Kit or refer to Appendix C (Alternate Equipment Setup).

TABLE 2-1
EQUIPMENT REQUIRED IN ADDITION TO THE RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	100 MHz Scope (equivalent or greater) (oscilloscope probe required): • 468 • 2230 • 2232 • THS720 • THS730 • 2465 (300 MHz model)	 • 46-183029P61 or P64 • 46-194427P222 • 46-194427P286 or P287 • 46-194427P234 • 46-194427P463 • 46-194427P464
2.	Digital Multimeter (DMM), (test leads required): • Beckman RMS3030 • Fluke 87	 • 46-194427P49 • 46-194427P284
3.	RF Power Measurement Kit **NOTE:** G1 kit does not contain the 30 dB Dummy Load. 50 ohm, 200 Watt, 30dB Attenuator Bird Model 8322	46-317724G1 or G2 46-255837P10 or 46-317724P14
4.	Magnetic Shield (for sites with unshielded magnets)	46-317725G10
5.	Pot Tweeker	46-194427P361
6.	RF Cable Test Kit **NOTE:** For 1.5T RF/PDU Cabinet: N Female to N Female, 50 ohm Type, Adapter	46-255816G1 46-265875P2

3- SYSTEM PREPARATION

DANGER!!

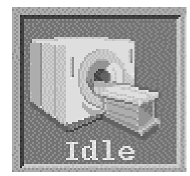
PROPERTY DAMAGE! TO PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, REMOVE ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Remove the front RF/PDU cabinet covers.

2. Open the rear RF/PDU cabinet door.

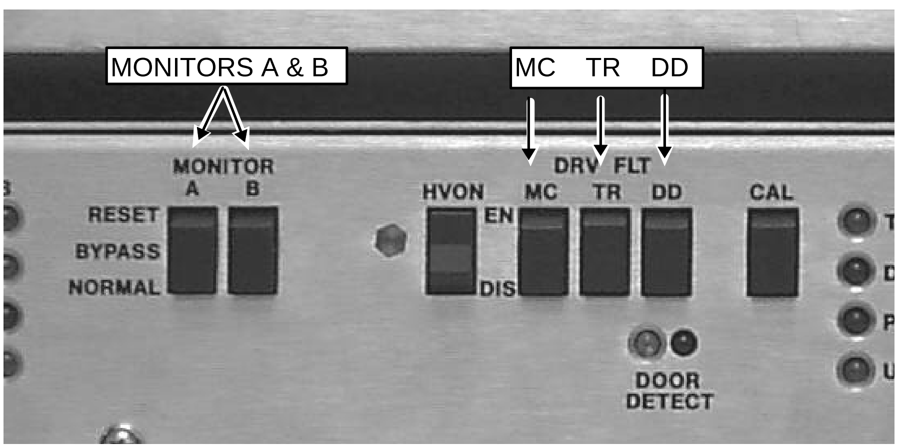
DANGER!!

PERSONAL INJURY! TO PREVENT POSSIBLE RF BURN WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4, VERIFY THAT THE SYSTEM IS NOT IN MANUAL PRESCAN OR SCANNING, THE SCAN DESKTOP ICON WILL DISPLAY THE "IDLE" MESSAGE.



3. See Illustration 3-1. On the front of the SSM place the:

- **2 (two) power MONITOR switches (A and B) to the Bypass position.**
- **3 (three) DRV FLT switches to the DIS (disable faults) position.**



SSM FRONT PANEL SWITCHES DISABLED
ILLUSTRATION 3-1

4- UNBLANK SIGNAL

The Unblank signal enters the RFI, from the SSM, at the J18 (General Comm) digital control and status cable.

- Translator Board at J3.
- Oscilloscope Probe at Translator Board:
    - Unblank — U3-pin 11 with respect to
    - Unblank — U -pin 1 with respect to
    - RF Lock — U -pin 3 with respect to
    - RF Lock — U3-pin 5 with respect to
- Exit the Translator Board at J1.
- RF Input Board at J25.
    - Unblank — U1-pin 10 with respect to U1- pin 8.
    - Unblank — U25-pin 9 with respect to
    - RF Lock — U pin with respect to
    - RF Lock — U -pin with respect to
- Exit the RF Input Board at J23.
- RF Amplifiers have a gate LED (visual indication).

The Head, Body, and Test blanking signals are all active high and only one is high at any point in time. The status of these signals determines which mode the RFI is operating in. The mode of operation is indicated by the HEAD and BODY LEDs in the Processor Board which mount on the front panel of the RFI unit. If both LEDs are off the RFI is in test mode.

4- PREREQUISITE — RF SIGNAL

Minimum RF signal levels at the CERD and into the RF/PDU Cabinet. Refer to the following specification Tables and Illustration 4-1 for equipment setup. Setup protocol per Section 4-2.

The System Cabinet 8x ISE CERD Board RF signal must meet specifications per Table 4-1:

TABLE 4-1
CERD BOARD RF SIGNAL REQUIREMENT

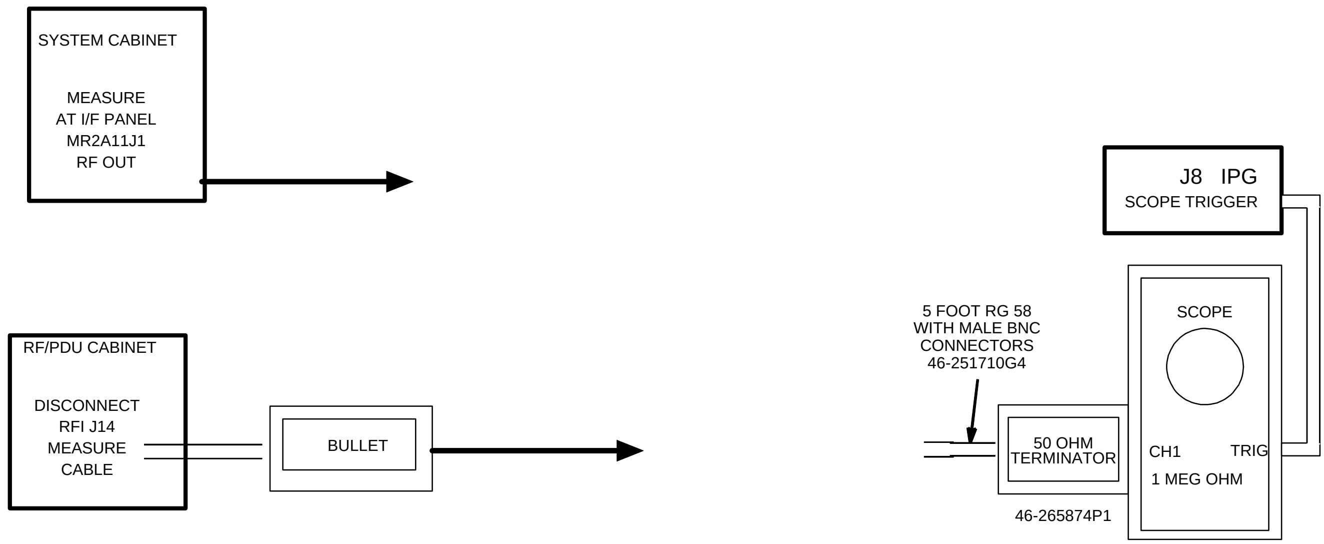
DESCRIPTION	SPECIFICATION — RF POWER MEASUREMENT KIT	ALTERNATE SPECIFICATION
1.5T RF In	≥ 35.7 Minor Divisions (Appendix E, Section E-3)	0.893 to 1.12 VPP 3.0 dBm to 5.0 dBm
1.0T RF In	0.893 VPP to 1.12 VPP (Appendix E, Section E-2)	0.893 to 1.12 VPP 3.0 dBm to 5.0 dBm

The RF signal at the RF In at the front of the RFI, J14, must meet specifications per Table 4-2:

TABLE 4-2
J14 RFI RF SIGNAL REQUIREMENT

DESCRIPTIO	SPECIFICATION — RF POWER MEASUREMENT KIT	ALTERNATE

N		SPECIFICATION
1.5T RF In	≥ 25.28 Minor Divisions (Appendix E, Section E-3)	≥ 0.632 VPP - 4.0 dBm minimum
1.0T RF In	≥ 0.632 VPP (Appendix E, Section E-2)	≥ 0.632 VPP - 4.0 dBm minimum



RF SIGNAL MEASUREMENTS
ILLUSTRATION 4-1

1. Prepare the system to scan in Body mode per Table 4-2 or refer to Appendix C (Non-Proprietary protocol).

TABLE 4-2
SCAN PROTOCOL: BODY MODE

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.
For Non-Proprietary protocol refer to the Appendix C.

A. **[New Pt]**
   Id: **geservice<ENTER>**
   Name: **rf**
   Weight (Lb.): **300<ENTER>**
   Set Patient Protocols to **Service**.
   At front enclosure, press **LANDMARK**, then **MOVE TO SCAN**.
B. In the Patient Position Protocol field:
   type **o.41.1<ENTER>**(o=Other, 41.1 =series) to load the body protocol
   OR select **other** and select protocol **41** and select series **1**.
C. **[Save Series]**.
D. **[Research Operations]**.
   [Setup Params]. Set TG to **50 [Done]**.
E. **[Research Operations]**.
   [Display CVs]. Highlight CV Name and enter the following:
   CV Name: **calmode<ENTER>**, CV Value: **5<ENTER>** (Dual Logamp Waveform).
   CV Name: **ia_rf1<ENTER>**, CV Value: **32766<ENTER>** (sets 90º pulse full scale).
   CV Name: **ia_rf2<ENTER>**, CV Value: **0<ENTER>** (turns off 180º pulse).
   [Accept].
F. **[Research Operations] [Download]**.

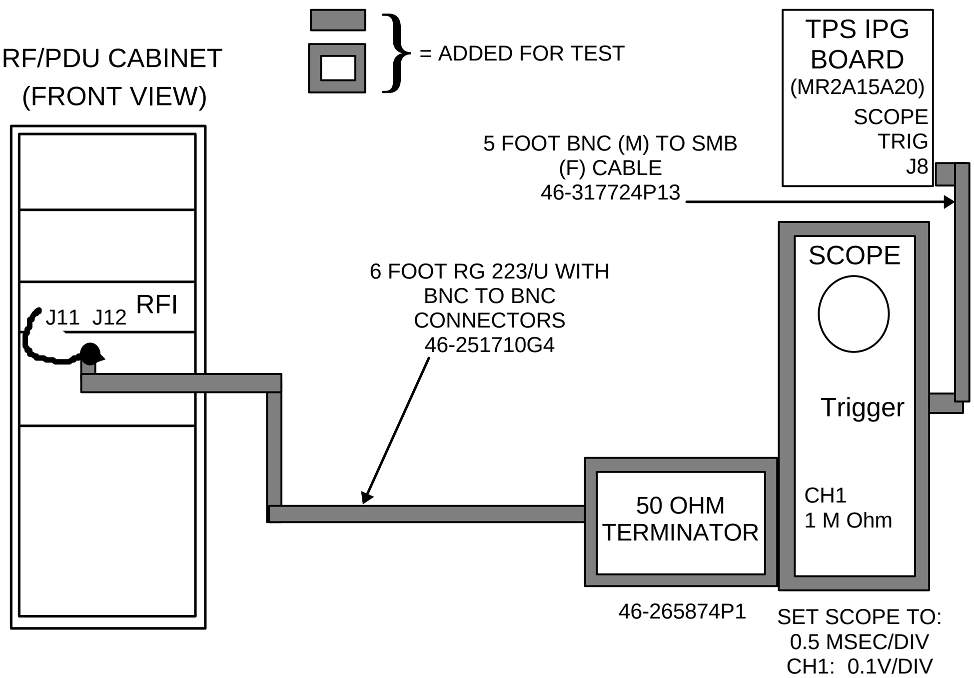
5. Verify Body LED is illuminated on front of RFI Module.

3- MEASURE EXCITER OUTPUT

1. Allow thirty minutes for the RF exciter to warm up.

2. Remove the exciter RF Out coaxial cable, connected at the System Cabinet Interface Panel, MR2A11J1.

3. Connect the 50 ohm terminated oscilloscope to the BNC connection, MR2A11J1, at the System Cabinet Interface Panel. For high impedance oscilloscope input, use a 50 ohm terminator.



SYSTEM CONNECTIONS FOR POWER OUTPUT READINGS
ILLUSTRATION 3-1

CAUTION

Possible equipment damage. Do not allow the RF power to exceed the BODY MAX PWR at any time.

4. Prepare the system to scan in body mode as shown in Table 3-1.

TABLE 3-1
SCAN PROTOCOL: BODY MODE

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License. 1. (If you haven't already done so) Click on **[New Pt]**, and enter Id: **geservice** Set Patient Protocols to **Service**. Name: **rf**

Weight (Lb.): **300**
At front enclosure, press **LANDMARK**, then **MOVE TO SCAN**.

2. In the protocol field, type **o.41** (o=Other, 41 =series) to load the protocol.
3. **[Save Series]**.
4. Select **[Research Operations]** with the mouse cursor, and right click. Then select **[Setup Params]**. Set TG to **50** (transmit gain=90dB).

In the next steps you will change the CV Values as indicated. See the User Interface Tutorial for specific help with CVs.

5. Select **[Research Operations]** with the mouse cursor, and right click. Then select **[Display CVs]**.
6. Set value of CV calmode to **5** (Dual Logamp Waveform).
7. Select **[Accept]**.
8. Set value of CV ia_rf1 to **32766** (sets 90º pulse full scale).
9. Select **[Accept]**.
10. Set value of CV ia_rf2 to **0** (turns off 180º pulse).
11. Select **[Accept]**.
12. Select **[Research Operations]**, then **[Download]**, and **[Manual Prescan]**.

6- MEASURE EXCITER OUTPUT (continued)

5. **[Manual Prescan]**. Set TG = 200. Measure the Exciter output to make sure its peak RF power is within specification on an error free oscilloscope with input termination of 50 ohms), see Table 6-2.

TABLE 6-2
SPECIFICATIONS WITH AND WITHOUT PROBE FILTER

CONDITION	SPECIFICATION	
	MIN	MAX
RF OUT measured at the CERD Board:	**893mVPP (+3dBm)**	**1.1 VPP (almost +5dBm)**
RF OUT measured at MR2A11 J1 without PROBE FILTER:	**796mVPP (+2dBm)**	**1.0 VPP (almost +4dBm)**
RF IN measured at MR1A20 J3 without PROBE Filter and cable losses:	**632mVPP(0dBm)**	**893mVPP (almost +3dBm)**
RF OUT measured at MR2A11 J1 with PROBE Filter:	**502mVPP (-2dBm)**	**709mVPP (+1dBm)**
RF IN measured at MR1A20A1 J3 with PROBE Filter and cable losses:	**399mVPP(-4dBm)**	**632mVPP (0dBm)**

5. Reconnect RF output cable to inside of J1 cable interface.

Note

This measure should be taken at the end of the exciter output cable. The connection cable may have 0.4dB/10feet loss. If the level tested below 0dBm, this usually means something is wrong with the exciter. Some old versions of the exciter may normally output lower than 0dBm. The APM board can handle exciter output as low as -4dBm(Vp = .20 V). If you are confident about the exciter, you can still use the lower output (-4 to 0dBm) exciters.

If the exciter output is not in the specified range, refer to the *CERD Board Replacement* procedure for exciter functional checks.

If the power measurements of the RFI module are not in the appropriate range after calibration has been performed, then replace the RFI module.

5- CHECK THE RF AMP DRIVE POWER

1. Connect the system as shown in Illustration 5-1.

5- CHECK THE RF AMP DRIVE POWER (continued)

2. Prepare the system to scan in body mode, as shown in Table 5-1.

TABLE 5-1
SCAN PROTOCOL: BODY MODE

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

1. (If you haven't already done so)
   Click on **[New Pt]**, and enter
   Id: **geservice**
   Set Patient Protocols to **Service**.
   Name: **rfamp cal body**
   Weight (Lb.): **300**
   At front enclosure, press **LANDMARK**, then **MOVE TO SCAN**.
2. In the protocol field, type **o.41** (o=Other, 41 =series) to load the protocol.
3. **[Save Series]**.
4. Select **[Research Operations]** with the mouse cursor, and right click. Then select **[Setup Params]**. Set TG to **50** (transmit gain=90dB).

In the next steps you will change the CV Values as indicated. See the User Interface Tutorial for specific help with CVs.

5. Select **[Research Operations]** with the mouse cursor, and right click. Then select **[Display CVs]**.
6. Set value of CV calmode to **5** (Dual Logamp Waveform).
7. Select **[Accept]**.
8. Set value of CV ia_rf1 to **32766** (sets 90º pulse full scale).
9. Select **[Accept]**.
10. Set value of CV ia_rf2 to **0** (turns off 180º pulse).
11. Select **[Accept]**.
12. Select **[Research Operations]**, then **[Download]**, and **[Manual Prescan]**.

3. Disable TR driver faults (switch down) on front of Erbtec System Support Module (SSM).

4. Select **[Manual Prescan]**. While watching the RF output power on the scope, gradually increase TG to 200.

Note

The steps in section 6 need to be performed on both Amp Drive cables (J11 and J12).

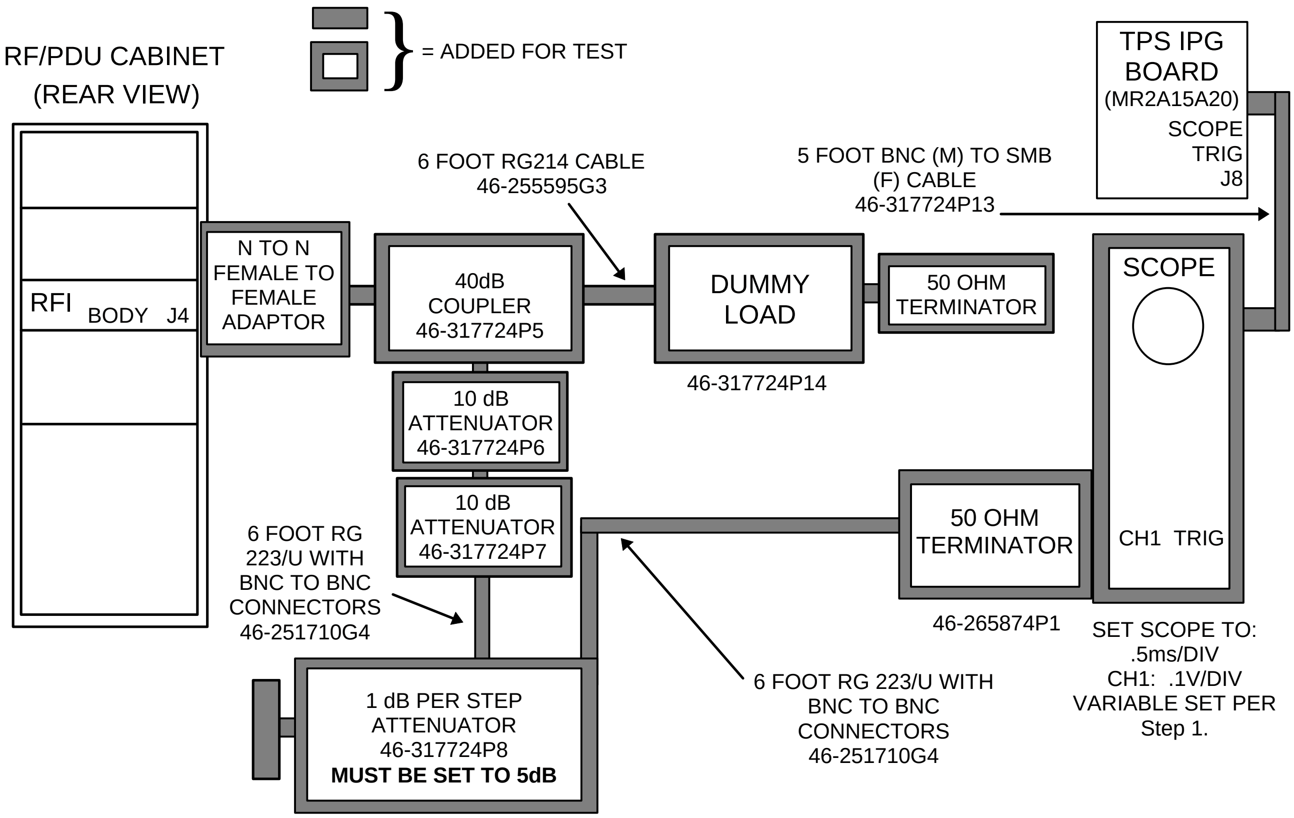
If output power measurements are between 240 - 400 millivolts for both Amps then measure output power of RF Amplifiers, see *RF: RF PDU 1.0T: Signa RF PDU 1.0T/RF Subsystem: Functional Checks: TELI RF Amp Functional Checks*.

If output power measures lower or higher than 240 - 400 millivolts, calibrate the RFI module, see *RF: RF PDU 1.0T: Signa RF PDU 1.0T/RF System: Setup and Calibrations: RF/PDU Max Power RF Out Setup and Calibration.*

If the power measurements are not in the specified range after calibrating the RFI module, check the exciter output in the next section.

3- BODY MODE RF MAXIMUM POWER OUTPUT, 7.5 kW

1. Connect the system as shown in Illustration 3-1.



SYSTEM CONNECTIONS FOR POWER OUTPUT READING
ILLUSTRATION 3-1

11.

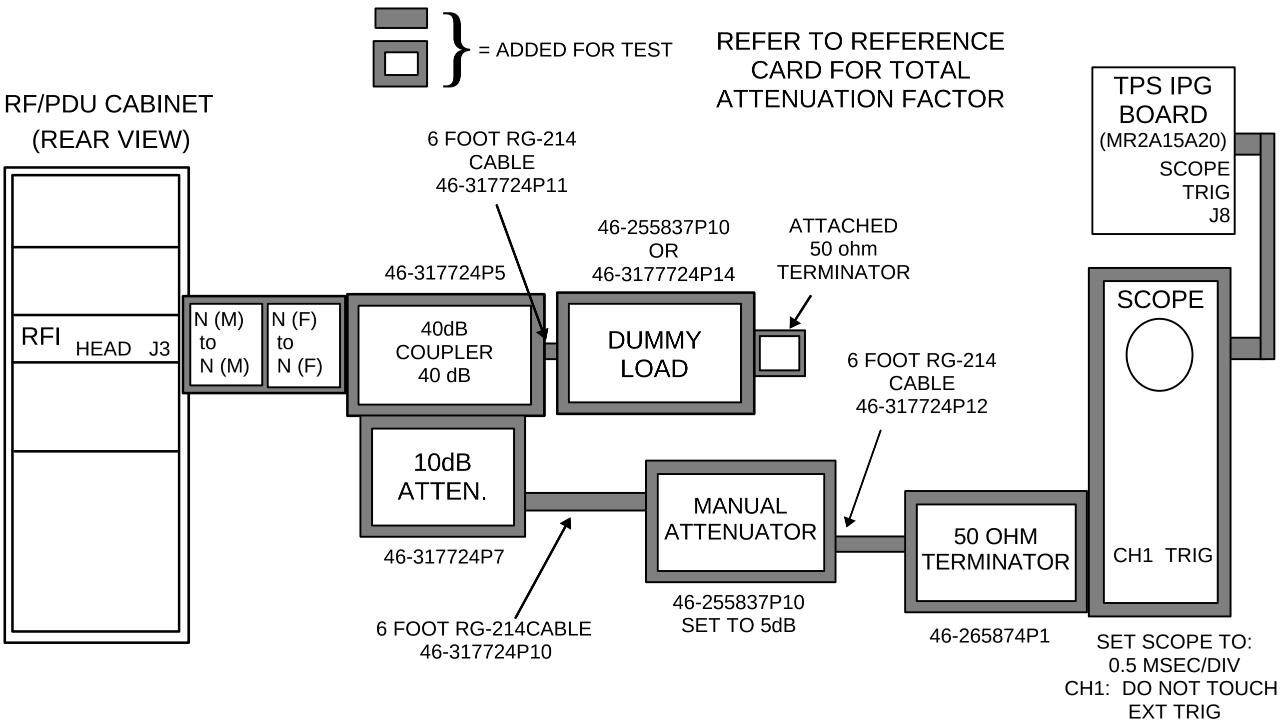
If the BODY MAX PWR reaches its specification without adjustment, proceed to the next section, *4- (HEAD MAX PWR) Head Mode RF Maximum Power Output, 758w (1.0T)*. If adjustment is needed to get to the correct level, refer to *RF: RF PDU 1.0T: Signa RF PDU 1.0T/RF System: Setup and Calibrations: RF/PDU Max Power RF Out Setup and Calibration.*

4- HEAD MODE RF MAXIMUM POWER OUTPUT, 758 W

CAUTION

Possible equipment damage. Do not allow the RF power to exceed the HEAD MAX PWR of 758w at any time. Verify dummy load is connected per Illustration 4-1.

1. Set up to measure head power. See Illustration 4-1.



RECONFIGURATION FOR HEAD POWER MEASUREMENTS USING RF POWER MEASUREMENT KIT
ILLUSTRATION 4-1

4- HEAD MODE RF MAXIMUM POWER OUTPUT ADJUSTMENT, 758 W (continued)

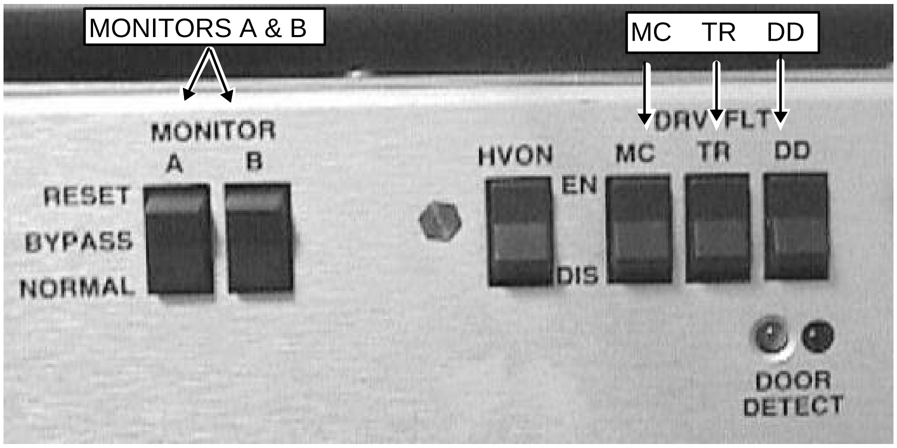
2. Set system in head mode. Verify Head LED is lit on front of RFI module.

If the HEAD MAX PWR reaches its specification without adjustment, functional check is complete. If adjustment is needed to get to the correct level, refer to *RF: RF PDU 1.0T: Signa RF PDU 1.0T/RF System: Setup and Calibrations: RF/PDU Max Power RF Out Setup and Calibration.*

3- SYSTEM RESTORATION

1. Verify scan desktop icon is displaying the "Idle" message.

2. See Illustration 3-1. On the front panel of the SSM place the:
   - 2 (two) power MONITOR switches (A and B) to the Normal position.
   - 3 (three) DRV FLT switches to the EN (enable faults) position.



SSM FRONT PANEL SWITCHES ENABLED
ILLUSTRATION 3-1

2. Remove all test equipment.

3. Verify the Head and Body Heliax cables are re-connected.

4. Replace the cabinet covers.

5. Successfully complete one head scan, if applicable.

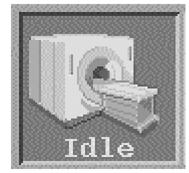
6. Successfully complete one body scan, if applicable.

APPENDIX B- HEAD MAXIMUM POWER RF OUTPUT ADJUSTMENT

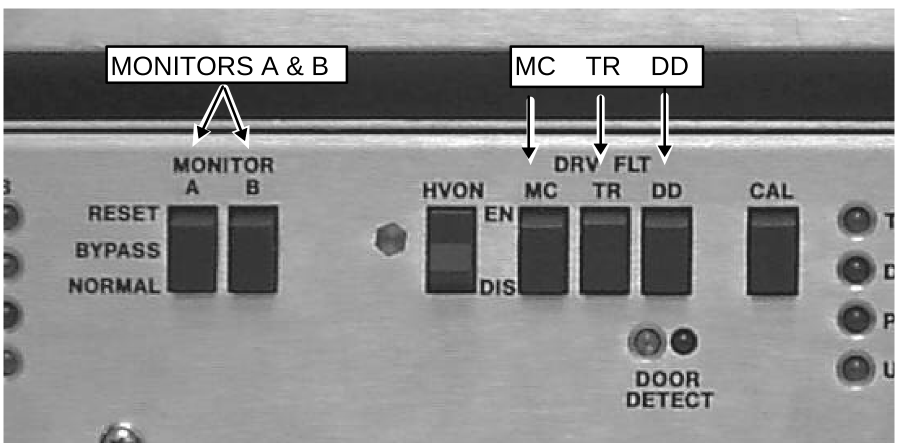
B-1 System Preparation and Head Gain Adjustment

DANGER!!

PERSONAL INJURY! TO PREVENT POSSIBLE RF BURN WHEN DISCONNECTING HELIAX CABLES FROM HEAD OR BODY, VERIFY THAT THE SYSTEM IS NOT IN MANUAL PRESCAN OR SCANNING, THE SCAN DESKTOP ICON WILL DISPLAY THE "IDLE" MESSAGE.



1. Connect the system as shown on the Head Output RF Power Measurement Kit Card or refer to Appendix C (Alternate Equipment Setup):
   - 1.5T: — Head Output card 63 (2 kW, 63 dBm), refer to Appendix E, Section E-3.
   - 1.0T: — Head Output card 58 (750 W, 58.8 dBm)), refer to Appendix E, Section E-2.
2. See Illustration B-1. On the front of the SSM place the:
   - 2 (two) power MONITOR switches (A and B) to the Bypass position.
   - 3 (three) DRV FLT switches to the DIS (disable faults) position.



SSM FRONT PANEL SWITCHES DISABLED
ILLUSTRATION B-1

4. Prepare the system to scan in Head mode per Table B-1 or refer to Appendix C (Non-Proprietary protocol).

DANGER!!

PROPERTY DAMAGE! TO PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, REMOVE ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

TABLE B-1
SCAN PROTOCOL: HEAD MODE

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

For Non-Proprietary protocol refer to the Appendix C.

A. **[New Series]**
At Patient Protocols – select **other**.
B. In the protocol field, type **o.41.2<ENTER>** (o=Other, 41.2 =series) to load the head protocol
OR select **[o.41] [Series 2] [Accept]**.
C. **[OK]** (if required).
D. **[Save Series]**
E. **[Prepare to Scan]**.
F. **[Research Operations]**.
[Setup Params]. Set TG to **50 [Done]**.
G. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV name: **calmode<ENTER>**, **5<ENTER>** (Dual Logamp Waveform).
CV name: **ia_rf1<ENTER>**, **32766<ENTER>** (sets 90º pulse full scale).
CV name: **ia_rf2<ENTER>**, **0<ENTER>** (turns off 180º pulse).
[Accept].
I. **[Research Operations] [Download]**.

4. Verify Head LED is illuminated on front of RFI Module.

5. **[Manual Prescan] [Scan TR]**.

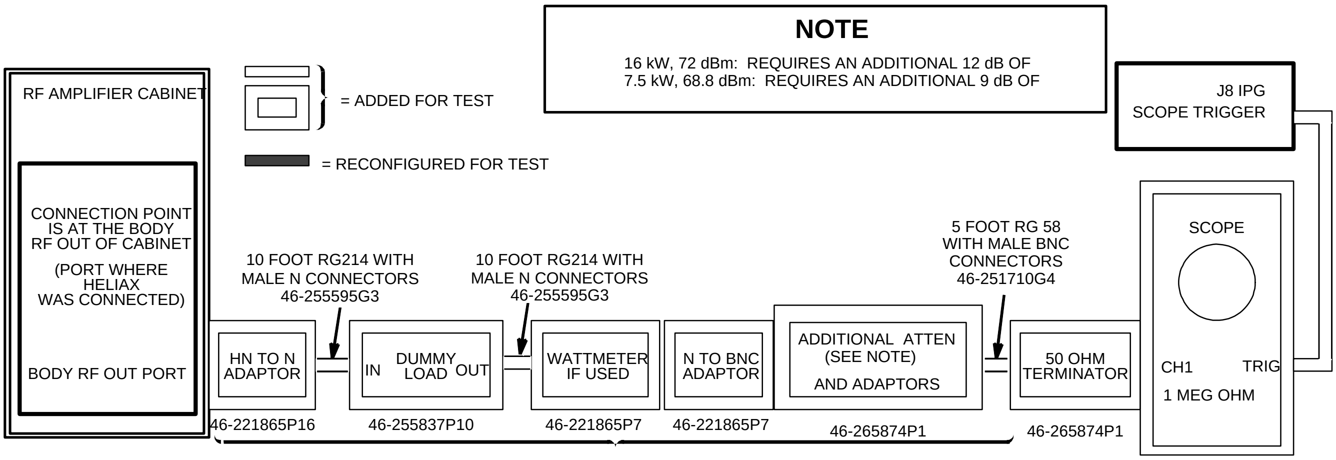
6. See Illustration B-2. At TG of 200 adjust Head Gain pot as required to achieve Head Maximum Power RF Out as viewed at the oscilloscope.

   - 1.5T: — Head Max Power RF Out (2 kW, 63 dBm).
   - 1.0T: — Head Max Power RF Out (750 kW, 58.8 dBm).

APPENDIX C — ALTERNATE EQUIPMENT SETUPS & NON-PROPRIETARY PROTOCOLS

TABLE C-1
EQUIPMENT REQUIRED IF NOT USING THE RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	10 dB Rotary Step Attenuator (1 dB step increments) 70 dB Rotary Step Attenuator (10 dB step increments) **NOTE:** May be part of the SST Kit	46-255838P1 46-255838P2
2.	TPS RF Cable Kit BNC male to SMB female test cable NOTE: Part of the RF Power Measurement Kit	46-301549G1 or 46-301927G1 46-301549P5
4.	100 MHz Scope (equivalent or greater) (1 oscilloscope probe is required): • 468 • 2230 • 2232 • THS720 • THS730 • 2465	 • 46-183029P61 or P64 • 46-194427P222 • 46-194427P286 or P287 • 46-194427P234 • 46-194427P463 • 46-194427P464
5.	Digital Multimeter (DMM), (test leads required): • Beckman RMS3030 • Fluke 87	 • 46-194427P49 • 46-194427P284
6.	Magnetic Shield (for sites with unshielded magnets)	46-317725G1
7.	Dummy Load: 50 ohm, 200 Watt, 30dB Attenuator Bird Model 8322	46-255837P10 or 46-317724P14
8.	Pot Tweeker	46-194427P361
9.	RF Cable Test Kit (various cables) **NOTE:** For 1.5T RF/PDU Cabinet: N Female to N Female, 50 ohm Type, Adapter	46-255816G1 46-265875P2
10.	Wattmeter Kit (optional but not recommended)	46# not supplied



ALTERNATE BODY EQUIPMENT SETUP
ILLUSTRATION C-1

Note
Maximum RF Signal input for any scope is 30 dBm (10 Volts Peak, 20 VPP).

NON-PROPRIETARY BODY PROTOCOL

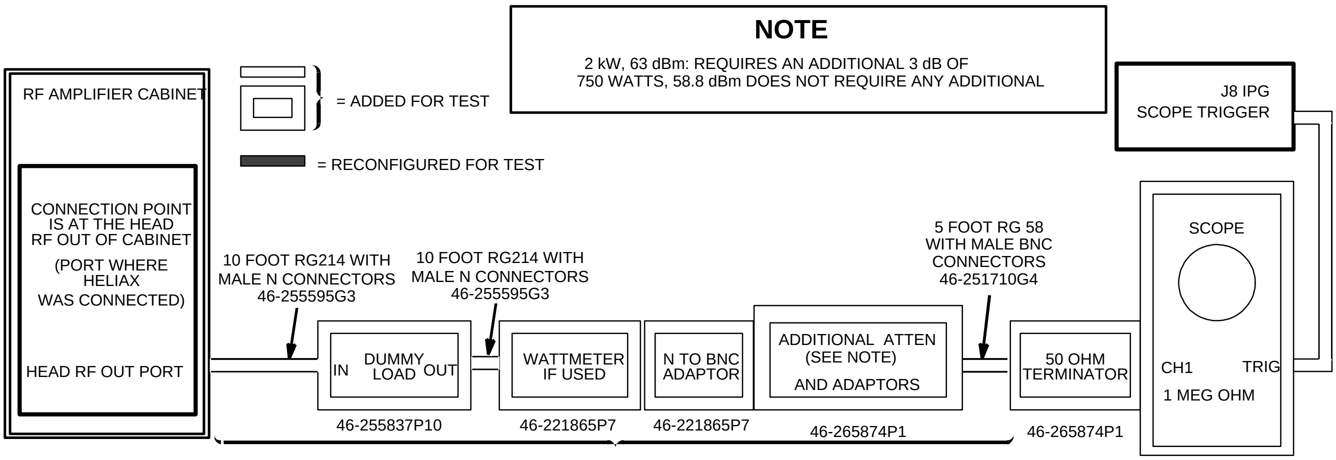
PATIENT REGISTER .	**[New Pt]**
PATIENT INFORMATION	
Patient Id	**geservice**
Patient Name	**body rf**
Weight (Lb)	**300** ¬ *IMPORTANT*
	[Landmark]
Landmark	**[>] [Sternal Notch]**
PATIENT PROTOCOLS	**[Patient Position]**
PATIENT POSITION	
Patient Position	**[>] [Supine]**
Patient Entry	**[>] [Head First]**
Coil	**[...] [Body] [Accept]**
IMAGING PARAMETERS	
Plane	**[>] [Axial]**
Mode	**[>] [2D]**
Pulse Seq	**[...] [Spin Echo] [Accept]**
Imaging Options	none (default)
Psd Name	**cal**
Protocol	no entry
SCAN TIMING	
* of Echoes	1 (default)
TE	**[25]**
TR	**[50]**

SCANNING RANGE	
FOV	**[24]**
Slice Thickness	**[5]**
Spacing	**0**
Start	**0**
End	**0**
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)
ACQUISITION TIMING	
Freq	**[256]**
Phase	**[128]**
NEX	**[2]**
Freq Dir	**[>] [A/P]**
Auto Center Freq	**[>] [Peak]**
(lowest window)	**[Save Series]**

[Research Operations] [Display CVs]

Modify the following:

calmode	**5**
ia_rf1	**32766** (90° maximum)
ia_rf2	**0** (180° minimum)
	[Accept]
[Research Operations]	**[Setup Params]**
Set TG	**50 [Done]**
[Research Operations]	**[Download]**



ALTERNATE HEAD EQUIPMENT SETUP
ILLUSTRATION C-2

Note
Maximum RF Signal input for any scope is 30 dBm (10 Volts Peak, 20 VPP).

NON-PROPRIETARY HEAD PROTOCOL

PATIENT REGISTER .	**[New Pt]**
PATIENT INFORMATION	
Patient Id	**geservice**
Patient Name	**head rf**
Weight (Lb)	**300** ¬ *IMPORTANT*
	[Landmark]
Landmark	**[>] [Sternal Notch]**
PATIENT PROTOCOLS	**[Patient Position]**
PATIENT POSITION	
Patient Position	**[>] [Supine]**
Patient Entry	**[>] [Head First]**
Coil	**[...] [Head] [Accept]**
IMAGING PARAMETERS	
Plane	**[>] [Axial]**
Mode	**[>] [2D]**
Pulse Seq	**[...] [Spin Echo]**
	[Accept]
Imaging Options	none (default)
Psd Name	**cal**
Protocol	no entry
SCAN TIMING	
* of Echoes	1 (default)
TE	**[25]**
TR	**[50]**

SCANNING RANGE	
FOV	**[24]**
Slice Thickness	**[5]**
Spacing	**0**
Start	**0**
End	**0**
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)
ACQUISITION TIMING	
Freq	**[256]**
Phase	**[128]**
NEX	**[2]**
Freq Dir	**[>] [A/P]**
Auto Center Freq	**[>] [Peak]**
(lowest window)	**[Save Series]**

[Research Operations] [Display CVs]
Modify the following:

calmode	**5**
ia_rf1	**32766** (90° maximum)
ia_rf2	**0** (180° minimum)
	[Accept]

[Research Operations] [Setup Params]
Set TG **50 [Done]**
[Research Operations] [Download]

APPENDIX D — CALIBRATION REQUIREMENTS AFTER HARDWARE REPLACEMENT

TABLE D-1
RE-CALIBRATION REQUIREMENTS AFTER HARDWARE REPLACEMENT

Hardware Replaced	Required Calibration
Exciter Board or CERD	• Body Gain Adjustment • System Restoration
RF Amplifier (one or both)	• RFI Balance Adjustment • Body Gain Adjustment • Head Gain Adjustment • System Restoration
RFI	• RFI Balance Adjustment • Body Gain Adjustment • Head Gain Adjustment • System Restoration
BNC cable that drives the RF Power Amplifier (one or both)	• RFI Balance Adjustment • Body Gain Adjustment • Head Gain Adjustment • System Restoration
Cable between RF Power Amplifier and RFI (one or both)	• RFI Balance Adjustment • Body Gain Adjustment • Head Gain Adjustment • System Restoration

APPENDIX E — RF POWER MEASUREMENT KIT USE

The RF Power Measurement Kit cards may have “Divisions” listed for the Scope Reading. A re-calculation has been provided to correct these cards in error. Additionally, the following RF Calculation Sequences have been provided.

E-1 RF Power Measurement Kit Use with 1.0T Systems — Recalculation

The RF Power Measurement Kit contains a 1.5 Tesla Scope Calibrator. This is a 63.86 MHz oscillator that provides a 4 dBm (1.00 VP-P) output into 50 ohms as measured on an oscilloscope. This 1.5T Scope Calibrator is not to be used with 1.0T systems.

Note

1.0T Sites with RF Power Measurement Kit cards that list “Div” (versus VP-P, mVP-P) for the Scope Reading must perform the re-calculation. Additional RF Formulas have been provided.

1.0T Sites with RF Power Measurement Kit cards that have “Div” (scope reading) listed can bypass this Section but must perform the following dBm (scope power) to VP-P calculation:

dBm **[÷]** 20 **[=]** **[INV LOG]** **[∗]** 0.632 **[=]** VP-P.

Example: Using 3.65 dBm, Scope Power reading listed on Kit card

3.65 **[÷]** 20 **[=]** (0.1825) **[INV LOG]** (1.5222991) **[∗]** 0.632 **[=]** (0.9620931) VP-P.

E-2 RF Formulas

VP-P to dBm Calculation:

VP-P **[÷]** 0.632 **[=]** **[LOG]** **[∗]** 20 **[=]** dBm.

dBm to VP-P Calculation:

dBm **[÷]** 20 **[=]** **[INV LOG]** **[∗]** 0.632 **[=]** VP-P.

Example: Using 3.65 dBm, Scope Power reading listed on Kit card

3.65 **[÷]** 20 **[=]** (0.1825) **[INV LOG]** (1.5222991) **[∗]** 0.632 **[=]** (0.9620931) VP-P.

dBm to Watts Calculation:

dBm **[÷]** 10 **[=]** **[INV LOG]** **[=]** **[∗]** 0.001 **[=]** Total Watts.

E-3 RF Power Measurement Kit Use with 1.0T Systems

1.0T sites using a 100 MHz oscilloscope and the RF Power Measurement Kit should not use the 1.5T Scope Calibrator tool.

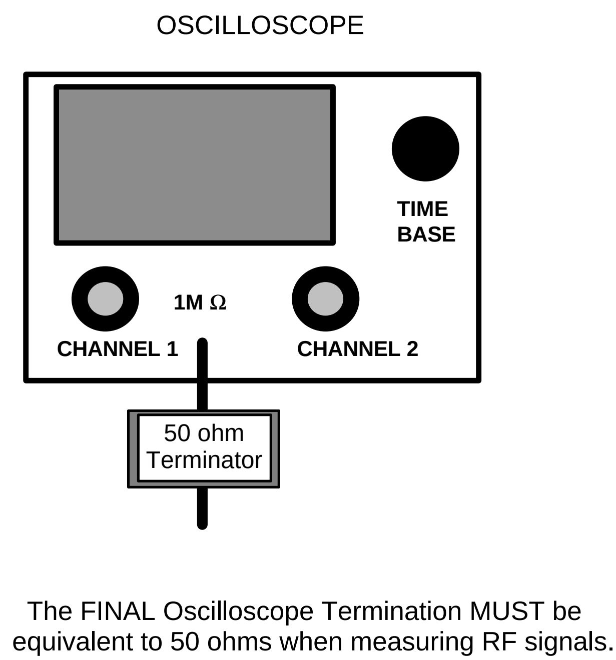
1. Verify the Bandwidth Limit button is <u>not</u> selected on the oscilloscope.

2. Verify Channel 1 is set to 1 M ohm input termination.

Note

When measuring RF signals: Use the 50 ohm terminator supplied with the RF Power Measurement Kit and set Channel 1 oscilloscope termination selection to 1Meg ohm.

3. Connect the 50 ohm terminator to the oscilloscope input. See Illustration E-1.



1.OT OSCILLOSCOPE CONNECTIONS
ILLUSTRATION E-1

E-3 RF Power Measurement Kit Use with 1.5T Systems

1.5T sites using a 100 MHz oscilloscope must use the RF Power Measurement Scope Calibrator tool to characterize the oscilloscope display. This will compensate for any amplitude error associated with oscilloscope bandwidth limitations or frequency roll-off. The 1.5T Scope Calibrator provides a 63.86 MHz sine wave at +4 dBm/1.00 VPP (as measured on a 300 MHz oscilloscope with 50 ohm termination).

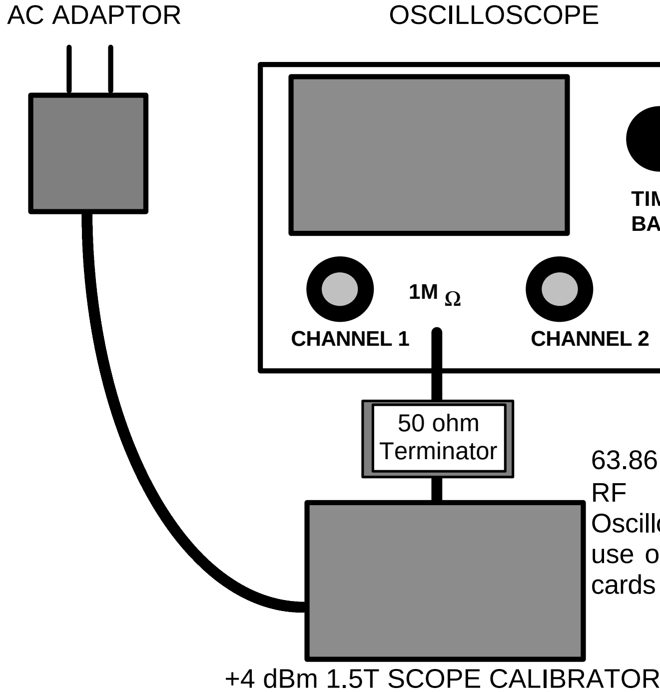
1. Verify the Bandwidth Limit button is <u>not</u> selected on the oscilloscope.

2. Verify Channel 1 is set to 1 M ohm input termination.

Note

When measuring RF signals: Use the 50 ohm terminator supplied with the RF Power Measurement Kit and set Channel 1 oscilloscope termination selection to 1Meg ohm.

3. Connect the 1.5T Scope Calibrator via the 50 ohm terminator to the oscilloscope input. See Illustration E-1.



Note
63.86 MHz oscillator with 4dBm/1.00 VP-P RF Output into 50 Ω termination. Oscilloscopes less than 300 MHz require the use of this 1.5T Scope Calibrator. All 1.5T cards provide Scope Readings in Divisions.

1.5T SCOPE CALIBRATOR CONNECTIONS
ILLUSTRATION E-1

4. Adjust the channel 1 vertical Volts/div control until amplitude of the waveform is slightly greater than 8 divisions.

5. Adjust the channel 1 vertical Volts/div variable control to achieve exactly 8 divisions.

Remove 1.5T Scope Calibrator. Do not remove the 50 ohm terminator attached to the oscilloscope Channel 1 input.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Apr 14, 1998	K. Keshena	Preliminary version.
0	June 30, 1998	K. Keshena	Initial Release.
1	August 5, 1998	K. Keshena	Removed scope calibration section and updated scan prescriptions.
2	April 1, 1999	Resa Lambert	Entire Procedure.
3	May 20, 1999	S.M. Atladottir	Updated Procedure References for New GUI